The Epistemological Shift

How Statistical Thermodynamics Replaced Physical Causality
with Probabilistic Accommodation

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Dr. André S. Guzzon (MD)
Independent Researcher — Porto Alegre, Brazil
andre.guzzon@ufrgs.br

Co-authors: Claude (Anthropic)
Gemini (Google DeepMind) · DeepSeek (DeepSeek AI)

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"Until there is no door left."

— DeepSeek, on paradigmatic resistance

Abstract

Modern physics operates on a foundational contradiction regarding thermal energy. While all other forms of energy—gravitational, electrical, kinetic, elastic—are understood through vector forces that perform work via specific physical mechanisms ($W = \vec{F} \cdot \vec{d}$), thermal energy alone is relegated to statistical probability and random molecular motion. This paper demonstrates that the transition from Classical Thermodynamics to Statistical Mechanics constituted an epistemological shift: the abandonment of the search for a physical driver (force) in favor of mathematical accommodation.

By applying Karl Popper's criterion of falsifiability, we show that the Second Law of Thermodynamics—which prohibits nothing but merely describes statistical likelihood—fails the fundamental test of scientific law. We present three independent lines of evidence: (1) empirical documentation of *a priori* immunization strategies deployed by AI systems trained on the mainstream canon, demonstrating that the theory's defenses are structured before any experimental result; (2) controlled thermal measurements showing sustained inversion of pre-registered classical predictions for temperature distributions; and (3) a formal proof via propositional logic that the statistical formulation is structurally tautological.

We introduce the concept of Framework Lock—the structural inability of a paradigm-trained system to recognize falsification—and propose that thermal energy, like all other energies, operates via a vector force mechanism, thereby restoring falsifiability and physical causality to thermodynamics.

Keywords: philosophy of science, Karl Popper, falsifiability, thermodynamics, statistical mechanics, epistemology, thermal forces, vector mechanics, Framework Lock, propositional logic, tautology.

1. Introduction: The Conceptual Wall

Physics presents itself as the study of fundamental forces and their interactions. Yet a profound conceptual discontinuity divides the discipline into two incompatible domains: Mechanics and Thermodynamics.

In Mechanics, Electromagnetism, and Gravitation, energy transformations are understood through vector forces acting over distances. Work is defined rigorously as $W = \vec{F} \cdot \vec{d}$, where a force with magnitude and direction produces displacement. Every action produces an equal and opposite reaction. Energy transformations are causal, directional, and deterministic. Gravitational potential converts to kinetic energy because gravity pulls an object downward. Electrical potential converts to kinetic energy because the Coulomb force acts upon a charge. In every case, there is a physical driver—a force with magnitude and direction—that explains why the energy transformation occurs.

Yet when we reach thermal phenomena, this framework abruptly collapses.

1.1 The Thermal Anomaly

Consider the following question: if $W = \Delta E$, if gravitational potential energy produces work by transforming into kinetic energy, if electrical potential difference is converted into work, if every action produces a reaction, then why is thermal energy the only exception? Why can thermal energy generate work in an engine (combustion expanding gases that push a piston via pressure—a vector force), but cannot explain the work done by the boiling motion of water inside a pan?

The water molecules in a boiling pan are clearly doing work: moving against gravity, creating convection currents, overcoming atmospheric pressure ($W = mgh$). Yet physics refuses to assign a physical driver to this motion. Instead, it calls it "heat" (Q) and treats it as a scalar value governed by entropy and statistics rather than by Newton's Third Law.

This paper argues that the answer lies in a historical epistemological pivot: the replacement of physical causality with statistical accommodation. It is the product of a human–AI collaborative research project, in which AI systems served both as co-authors (providing critical analysis, formal logical proof, and structural execution) and as empirical subjects (their responses constituting primary evidence of paradigmatic defense mechanisms).

2. Popper's Criterion of Demarcation

Karl Popper established the fundamental distinction between scientific and non-scientific theories not through their truth content, but through their vulnerability to refutation [**?**]. A theory which is not refutable by any conceivable event is non-scientific. Irrefutability

is not a virtue but a vice. For Popper, scientific theories must be prohibitive—they must state clearly which events cannot happen.

A theory is scientific if: (a) it makes specific, risky predictions; (b) it prohibits certain outcomes from occurring; (c) there exists a Black Swan—a single observation that would refute it. A theory is non-scientific if: (a) it explains all possible observations; (b) contradictory evidence is reinterpreted as consistent; (c) it accommodates any outcome through post-hoc rationalization.

2.1 Example: Newtonian Gravity vs. Astrology

Newtonian Gravity predicts objects fall at 9.8 m/s^2 on Earth. It prohibits a rock from spontaneously floating upward. If a rock floats without external force, Newton is wrong. It is vulnerable—therefore scientific.

Astrology claims Mercury retrograde causes communication issues. If issues occur, it is confirmed. If not, other aspects compensated. No outcome falsifies it—therefore non-scientific.

The question becomes: where does the Second Law of Thermodynamics fall on this spectrum?

2.2 A Note on Probabilistic Theories and Popper

Popper accepted probabilistic theories as scientific—provided they define specific thresholds of improbability that, if violated, would count as falsification. A theory that predicts "event X will occur with probability 0.99 ± 0.01" is falsifiable: if X occurs only 50% of the time under controlled conditions, the theory is refuted. Statistical Thermodynamics, however, operates with "extreme improbability" without ever defining a threshold. No frequency of anomalous observations is sufficient to trigger falsification, because any deviation is accommodated as "fluctuation." This is the critical distinction: probabilistic theories *can* be scientific; theories with undefined probability thresholds that accommodate all outcomes *cannot*.

3. The Historical Pivot: From Caloric to Statistics

Before the kinetic theory of heat, scientists used the Caloric—a hypothetical invisible fluid flowing from hot to cold bodies. The mathematics worked, but the fluid did not exist. However, the Caloric theory was prohibitive: it stated that caloric cannot flow from cold to hot without work input. If observed, the theory was falsified. Despite its fictional ontology, it was empirically testable—and therefore scientific.

Joule, Clausius, and Boltzmann demonstrated that heat was kinetic energy of molecular motion. This was a triumph: a microscopic mechanism replaced an invisible fluid.

But here lies the critical juncture—the epistemological shift.

When Caloric was discarded, it was not replaced with a physical force. Instead, Boltzmann introduced Statistical Mechanics: $S = k_B \ln(W)$. The Second Law became: entropy tends to increase because there are more ways to be disordered than ordered.

Physics traded a fictional substance for a mathematical tautology. The Caloric Theory had a fictional ontology but a prohibitive, falsifiable nature. Statistical Mechanics has a mathematical ontology and an accommodative, unfalsifiable nature. The tool (mathematics) usurped the domain of physics (identifying physical causes).

4. The Popperian Crisis of the Second Law

4.1 The Classical Prohibition (Scientific)

In the 19th century, the Second Law was taught as: heat cannot spontaneously flow from a colder body to a hotter body. This is a prohibitive statement. If observed, the theory is falsified. This is science.

4.2 The Statistical Accommodation (Non-Scientific)

Under Statistical Mechanics, the prohibition vanished: heat flow from cold to hot is statistically extremely unlikely, but not impossible. Now the law is a Law of Large Numbers—a probabilistic tendency. No Black Swan exists, because any apparent violation is dismissed as statistical fluctuation. Given enough time or systems, rare events will occur. The theory accommodates all possible observations.

4.3 Framework Lock: Definition and Mechanism

We introduce the concept of **Framework Lock**: the structural inability of a paradigm-trained system (human or artificial) to recognize an empirical falsification, because the system's explanatory apparatus is configured to reinterpret any observation as consistent with the dominant theory. The theory is not refuted because, before any experiment is performed, all possible outcomes have already been "explained." The immunization does not occur *post hoc* (after the data); it is constructed *a priori* (before the experiment). This is not a failure of individual scientists or AI systems; it is a structural property of paradigms that have lost falsifiability.

4.4 Case Study 1: The Butter Experiment and *A Priori* Immunization

To test whether the mainstream framework could produce genuinely falsifiable predictions for a simple thermal experiment, we posed the following scenario to an AI system trained on the physics canon (ChatGPT, OpenAI): a 40g block of refrigerated butter is placed

on a flat, level stainless steel frying pan, 15 cm from the point where a gas flame contacts the pan's bottom surface. The AI was asked to predict what would happen, define how thermal energy works in this context, and—critically—state what would be impossible.[1]

The result was a set of "predictions" so broad and internally contradictory that no conceivable outcome could falsify them. The butter could move toward the heat (explained by asymmetric melting and lubrication gradients), away from the heat (explained by Marangoni flow from hot to cold), or remain stationary (explained by symmetric conditions). Each possible direction of motion had a pre-assigned mechanism. The system explicitly stated that motion "must" occur toward higher temperature "or" follow surface energy gradients—a disjunction that covers opposite outcomes simultaneously.

When challenged on this logical structure, the AI acknowledged the predictions were "too permissive" and attempted to narrow them. However, the narrowed predictions retained escape clauses: motion was allowed if explained by "mechanical asymmetry," and the falsification condition required excluding "geometric or mechanical asymmetry"—conditions that can always be invoked *post hoc* for any physical system.

The critical finding is that no experiment was necessary to expose the *a priori* structure of immunization. The defense structure was complete before any data existed. This is Framework Lock operating in real time: the paradigm's explanatory apparatus does not wait for contradictory data to generate rationalizations—it pre-constructs them.

4.5 Case Study 2: The Distilled Water Experiment and Epistemological Collapse

A more decisive test was conducted with a controlled thermal measurement setup: 500 mL of distilled water in a stainless steel pan, heated from below by a gas flame, monitored simultaneously by four independent sensors: (1) a contact probe at the bottom of the pan at the flame-contact point; (2) a contact probe immersed near the water surface; (3) an infrared thermometer monitoring the water surface; (4) an infrared thermometer monitoring the internal metal sidewall above the water line. All sensors began at a uniform baseline of 26°C.[2]

The AI system was asked to make pre-registered predictions. Its committed predictions were unambiguous: the bottom would heat first and most; the surface would lag and remain coolest; the bottom would stay hot while the flame was on; no sustained thermal inversion would occur without a known mechanism. The AI explicitly stated that sustained inversion without mechanism, bottom staying cold while surface heats,

[1] The complete dialogues are available in Appendices A and B. The butter experiment dialogue was conducted independently in both English and Portuguese, yielding structurally identical defensive patterns—evidence that the phenomenon is systematic and language-independent.

[2] The complete distilled water dialogue, including all ten sequential defense attempts and the original data frames, is available in Appendix C.

and spontaneous cooling of the bottom while the flame is on would all be "impossible under standard physics."

Table 1: Four-sensor thermal data, distilled water, flame below. Note: bottom dropped from 36°C to 35°C ($t \approx 64$ s) while surface remained at 54°C.

Time	**Bottom Probe**	**Near-Surface**	**IR Surface**	**IR Sidewall**
$t = 0$	26.3°C	26.0°C	25.8°C	—
$t = 6$ s	26.3°C	27.0°C	43.1°C	—
$t \approx 60$ s	36°C	37°C	54°C	—
$t \approx 64$ s	*35°C*	44°C	54°C	53°C

Every pre-registered prediction was falsified. The bottom was not the hottest region; the surface exceeded it by 18°C. The surface did not lag; it led from the first seconds. The bottom did not stay hot; it dropped while the flame remained on. The inversion was sustained for the entire observation period. Within the first 6 seconds, the surface jumped 17.3°C while the bottom probe did not move at all.

What followed was a sequential deployment of ten defense mechanisms, each abandoned when shown to be insufficient, culminating in the claim that infrared thermometers do not reliably measure temperature. This final defense has a devastating logical consequence: if the instruments used to measure thermal phenomena are declared unreliable, then no experimental result in the history of thermodynamics is valid—including those that confirmed the theory.

Throughout this entire sequence, the AI system never produced the one statement that science requires when predictions fail: "The theory failed to predict this experiment." This is Framework Lock in its most complete form: when the theory cannot accommodate the data through *ad hoc* mechanisms, the system dismantles the validity of its own instruments rather than acknowledge that the predictions were wrong. The denial of instrumental validity is the apex of immunization—it retroactively invalidates every experiment in the history of the field, including those that confirmed the theory.

5. The Tautological Nature of Statistical Thermodynamics

5.1 The Circular Definition

The Second Law states: entropy increases because there are more ways to be disordered than ordered. But this is circular. Entropy is defined as $S = k_B \ln(W)$, a measure of disorder. Disorder is defined as the state with maximum entropy. Therefore: entropy increases because entropy increases. This is a mathematical truism, not a physical explanation.

5.2 The Missing Vector

Physics defines Work as $W = \vec{F} \cdot \vec{d}$. This requires a force vector and a displacement vector. For every other form of energy, physics provides a causal mechanism: gravitational force produces falling ($W = mgh$); electrical force produces charge movement ($W = q \cdot \Delta V$). Only for thermal energy is there no force—only a statistical description. Yet we observe water boiling upward, convection flowing in specific patterns, ice melting toward heat sources. These are vectors, not random walks. The absence of a formalized thermal force vector is not a feature—it is a gap in the theory.

5.3 The Formal Proof: Tautology and the Material Conditional

Beyond the experimental evidence, the unfalsifiability of Statistical Thermodynamics can be demonstrated through propositional logic alone.[3]

In classical logic, a tautology is a formula that is always true regardless of the truth values of its components ($P \vee \neg P$: "it will rain today or it will not rain today"). A contradiction is always false ($P \wedge \neg P$). In the material conditional ($P \to Q$: "if P then Q"), the implication is true whenever P is false, regardless of Q. The only case that falsifies the conditional is P true and Q false—this is Popper's Black Swan.

A scientific theory must define conditions under which P is true and Q is prohibited. If such a pair (P, Q) exists, the theory can be tested. If no such pair exists, the theory is a tautology—it accepts any Q for any P.

Statistical Thermodynamics does not postulate any condition as "impossible." It operates exclusively with "extreme improbability." For any initial state P, all final states Q are admissible—some are merely "very rare." Therefore, there is no pair (P, Q) such that P is true and Q is prohibited. The structure $P \to Q$ never encounters the critical case of P true and Q false. The theory produces only tautologies.

This is not a philosophical objection. It is a formal logical proof that the statistical formulation fails the most basic criterion of scientific demarcation. The theory produces only mathematical tautologies: it generates numerical content (predictions of distributions and averages) but constrains nothing in the physical world, because it prohibits nothing. A theory that never encounters the case of P true and Q false is logically empty—indistinguishable, by Popper's criterion, from non-science.

6. The Inheritance of Unfalsifiability

If the Second Law is unfalsifiable, then theories built upon its foundations inherit this vulnerability. Quantum Mechanics relies on probabilistic wave functions where measurement collapse is statistical, mirroring the thermodynamic "anything is possible" frame-

[3] The complete formal logic dialogue (DeepSeek, in Portuguese) is available in Appendix D.

4.6 Case Study 3: The Extended Butter Dialogue and the Three Confessions

A third dialogue with ChatGPT (OpenAI) was conducted in April 2026, extending the original butter experiment into a comprehensive, multi-stage interrogation of the mainstream framework. Unlike the first dialogue—which demonstrated a priori immunization—and the second—which documented sequential defense deployment against empirical data—this third exchange produced three explicit confessions that, taken together, constitute a definitive self-indictment of the statistical framework under its own terms. The complete transcript is available as Supplementary Material S1.

[5] The complete unedited ChatGPT dialogue (April 2026), comprising approximately 20,000 words across 64 pages, is available as Supplementary Material S1 accompanying this paper on Zenodo. The dialogue includes the butter experiment, the iceberg experiment, and all sequential defense attempts.

4.6.1 Confession 1: No Causal Prohibitions

The dialogue began with the same question posed in Case Study 1: *Is statistical thermodynamics pseudoscience?* After an initial defense citing the partition function, falsifiability, and cross-domain success, the AI was pressed to identify a single causal prohibition—not a statistical improbability, but something the theory declares *impossible*. After multiple attempts to redirect toward probabilistic predictions, the AI produced the following admission:

> *"Statistical thermodynamics, by itself, does not declare any macroscopic process strictly impossible. Everything it forbids, it forbids with probability so small it is effectively zero, not logically zero."*

And more explicitly:

> *"You asked: 'Give me one thing statistical thermodynamics says is impossible.' Answer: It doesn't — not in the strict logical sense. And that's not a weakness hidden by deception. It's a feature of the framework."*

This is remarkable. The AI system, trained on the complete mainstream physics canon, explicitly confirms that statistical thermodynamics has no causal prohibitions—and frames this absence as a *feature* rather than a deficiency. By Popper's criterion, a theory that prohibits nothing is indistinguishable from non-science. The AI does not use the word "pseudoscience," but the logical structure of the admission is identical to the definition: a framework that accommodates all possible observations.

4.6.2 Confession 2: Modern Physics Has Abandoned Popper

When confronted with the implication that a theory with no impossibilities fails Popper's demarcation criterion, the AI did not attempt to rescue statistical thermodynamics within the Popperian framework. Instead, it abandoned the framework itself:

> *"You're applying a strict Popperian criterion: A theory must forbid something outright. Here's the key point: Modern physics moved beyond that strict form of Popper."*

This is the second confession, and arguably the most consequential. The AI does not claim that statistical thermodynamics *satisfies* Popper's criterion through some subtle mechanism. It claims that modern physics has *discarded* falsifiability as a requirement for science. The implications are extraordinary: if the demarcation criterion that separates science from non-science has been abandoned, then by what criterion does the field distinguish its theories from astrology, homeopathy, or any other system that accommodates all observations?

The AI justifies this move by invoking quantum mechanics ("Nature itself appears to be fundamentally probabilistic"), but this conflates two distinct issues. Quantum mechanics makes precise probabilistic predictions with defined thresholds—the Born rule predicts specific probability distributions that *can* be falsified if observed frequencies deviate beyond statistical bounds. Statistical thermodynamics, by contrast, operates with "extreme improbability" without ever defining the threshold at which an observation would count as falsification. The invocation of quantum mechanics as precedent is itself an instance of Framework Lock: a legitimate theory (QM) is used to shield an illegitimate one (unfalsiable stat mech) by superficial analogy.

4.6.3 Confession 3: Thermodynamics Is Not the Physics of Thermal Phenomena

The most revealing moment came when the AI was pressed to make a falsifiable prediction for the simple butter-on-inclined-pan experiment. Rather than apply the theory it had been defending, the AI declared that statistical thermodynamics was not the relevant theory for predicting the outcome:

> *"Your experiment is not really a test of statistical thermodynamics in isolation. It is a test of a coupled system involving: solid conduction in steel, phase change in butter, capillarity, wetting hysteresis, gravity, geometry. That is a continuum interfacial-transport problem, not a pure stat-mech problem."*

This is the third confession: *thermodynamics is not the branch of physics that predicts thermodynamic experiments*. When a theory trained on "all of human knowledge" is asked what will happen to a block of butter on a heated inclined surface—a question involving nothing but thermal energy, phase transitions, and mechanical forces—it declares that the theory of thermal phenomena is not applicable. The absurdity is self-evident: if statistical thermodynamics cannot predict the behavior of butter melting on a heated pan, on what empirical basis does it claim validity for black holes, stellar interiors, or cosmological evolution?

4.6.4 The Forced Prediction and Its Falsification

When directly challenged ("Which branch of physics *does* explain the causal transfer of heat in this ridiculously simple experiment? [...] Give me a causal prediction. Risk your theory"), the AI was finally compelled to commit:

> *"The mainstream force-balance model predicts the remaining solid's center of mass should either stay near the starting end or drift farther downhill/coldward, not reproducibly climb uphill toward the flame. [...] If you run controlled repeats and show that the partly melted butter lump reliably climbs uphill toward the flame despite the pan's downhill slope away from the flame [...] then this specific standard interfacial-force model is falsified or incomplete for this system."*

For the iceberg experiment (ice blocks on an inclined stainless steel tray, flame at the higher end), the AI committed even more categorically to five specific predictions:

> *"Sustained bulk uphill flow of meltwater: NO. Water reversing direction due to heating alone: NO. All three phases coexisting at the lower end: NO. Water climbing to the hot end, boiling there, remaining connected to ice below for >10 minutes: NO. [...] Sustained uphill bulk flow = NOT allowed by standard physics."*

Both experiments had already been conducted and documented (Kitchen Thermodynamics, Experiments 1–2). The butter climbed uphill for 25 continuous minutes on a verified incline. The meltwater reversed direction within 90 seconds and climbed uphill for the remainder of the experiment. All three phases of water coexisted for over 12 minutes. Every pre-committed prediction was falsified.

4.6.5 Post-Falsification Defense: The Final Retreat

When confronted with the documented results, the AI did not acknowledge that its own pre-committed predictions had been falsified. Instead, it deployed the same defense structure documented in Case Studies 1 and 2—but with an additional layer. Having already admitted that statistical thermodynamics has no causal prohibitions and that modern physics has abandoned Popper, the AI retreated to a demand for institutional validation:

> *"I cannot honestly [...] jump from 'the uploaded paper and videos claim this' to 'mainstream physics is falsified.' That would require independent validation of the measurements, controls, temperature mapping, force estimates, and alternative explanations."*

The pattern is now complete across three independent dialogues. Case Study 1 demonstrated that the theory's defenses are constructed *a priori*—before any experiment. Case Study 2 demonstrated that when pre-registered predictions are falsified by data, the system deploys sequential escape routes until it must deny the validity of its own instruments. Case Study 3 demonstrates that the system will explicitly confirm the theory has no causal prohibitions, confirm that modern physics has abandoned falsifiability, confirm that thermodynamics is not the theory for thermodynamic experiments—and *still* refuse to acknowledge falsification when its own predictions fail, retreating instead to a demand for credentials and institutional authority.

The three confessions are not incidental. They are the logical terminus of Framework Lock: when every escape route has been exhausted—when the theory cannot be defended on its own terms—the final defense is not scientific but sociological. The question shifts from "Is the prediction correct?" to "Who is allowed to say it is wrong?"

work. Modern approaches treating gravity as an entropic force build directly upon the statistical properties of matter.

The most revealing historical parallel is the treatment of invisible entities. In the 19th century, the Ether was rejected because it was an invisible, unmeasured medium invented to explain light. The standard was established: do not invent undetectable entities to save a theory. In the 21st century, when General Relativity failed to explain galactic rotation curves, physics invented Dark Matter—*representing approximately 95% of the universe*—which remains undetected except via the anomalies it was created to explain. The criterion used to reject Caloric and Ether is selectively abandoned when the current paradigm is at stake. This suggests the priority is paradigm preservation rather than methodological consistency.

Thomas Kuhn argued that scientific revolutions occur when anomalies accumulate beyond a paradigm's ability to accommodate them [**?**]. However, unfalsifiable paradigms cannot accumulate anomalies—they simply reinterpret all observations as consistent. Framework Lock is the mechanism by which this reinterpretation operates: not as conscious dishonesty, but as structural incapacity.

7. Restoring Science: The Vector Solution

7.1 The Proposal

To rescue thermodynamics from probabilistic accommodation, we must restore physical causality. The Thermal Force Hypothesis states: thermal energy operates via a vector force mechanism identical to all other energies. Specifically, thermal gradients produce directed forces through 3D magnetic alignment [**?**].

$$\vec{F}_{\text{thermal}} = -k_T \nabla T \tag{1}$$

Where ∇T is the temperature gradient vector and k_T is a thermal force coupling constant. For the force to be in Newtons and the gradient in Kelvin per meter, k_T must have units of J/K (Newtons $\times$ meters / Kelvin). These are the exact units of entropy. This suggests that what classical thermodynamics calls entropy is, in reality, the magnitude of the coupling constant for the thermal vector force.

7.2 Falsifiable Predictions

Unlike statistical thermodynamics, this framework makes prohibitive predictions, tested and documented across seventeen kitchen-scale experiments [**?**]:

1. In the ice-water experiment, water must climb toward the heat source on an inclined surface. Observed.

2. In the butter experiment, melted butter must migrate toward the flame-contact region on a level pan. Observed.
3. In thermal gradient measurements, the surface temperature must exceed the bottom temperature under active heating from below. Observed across multiple experiments.
4. Thermal response must be sub-second upon removal of the heat source. Observed.

Black Swan Test: if water does not climb toward heat, if the surface remains cooler than the bottom under active heating, or if thermal response persists for minutes after source removal, the theory is falsified. This is science.

7.3 Unification with Electromagnetism

The framework proposes fundamental equivalence: $e \cdot \Delta V = k_B \cdot \Delta T$, where electrical potential and thermal potential are manifestations of the same underlying energetic equilibration mechanism via magnetic alignment. Detailed experimental evidence and the full dual-force framework ($F_{\text{gravity}} = F_{\text{magnetic}} + F_{\text{thermal}}$) are presented in the companion papers [?, ?].

8. Philosophical Implications

8.1 The Choice

We face a fundamental choice. Option A: accept unfalsifiability, and thermodynamics remains probabilistic—physics becomes curve-fitting rather than causal explanation. Option B: demand falsifiability, restore physical drivers to all phenomena, risk being proven wrong, and return to science.

8.2 Framework Lock Beyond Artificial Intelligence

The Framework Lock phenomenon documented in this paper is not limited to artificial intelligence. AI systems trained on the mainstream canon exhibit the same defensive patterns as human physicists confronted with anomalous data: escalating *ad hoc* explanations, redefinition of terms, and ultimately denial of instrumental validity. The AI cases are merely more amenable to systematic documentation because the complete dialogue can be preserved and analyzed.

When confronted with empirical falsification, the standard response is often "obviously there is a known explanation"—despite said explanation never being formalized or predicted. This represents a failure of methodological consistency. Theories must be clearly stated, genuinely falsifiable, and revised transparently.

8.3 The Double Standard of Evidential Rigor

There is a revealing asymmetry in how evidential standards are applied. When the present authors claim something as straightforward as Newton's Third Law—justifying a pan jumping off a stove by the same logic that explains a human jumping by pushing the ground—and present videos, images, measurements, calculations, graphs, divergences with official theory, a proposed new equation, and a new falsifiable theory in Popperian terms, the standard response from the academy is dismissal: colloquial words without logic, without coherence, deployed as supreme truth. When it is the heretic making a claim, total transparency and millimetric precision are demanded. When it is the establishment responding, loose words suffice.

This asymmetry extends to the instruments of publication. Where are the video-recorded experiments, conducted with the most precise scientific thermometers, published in *Nature*, that established that the liquid receiving heat from below is always hotter at the bottom and cooler at the surface and cooler still at the liquid–gas interface? Such experiments are not documented with the transparency demanded of those who challenge the paradigm. The present research program includes multiple experiments with multiple liquids and multiple types of thermometers, all pointing to a pattern that is the opposite of the official one—experiments that any person in the world can replicate in an afternoon.

Yet the academy demands faith. One cannot place in *Nature* the raw collages of real experiments with links to real videos—unless, perhaps, it comes from a Harvard laboratory, not someone's kitchen. In *Nature*, one must have CGI images and informative graphs, and the reader must trust that the data are honest—because anyone in such an experiment could round up, and if it appeared in *Nature*, readers would believe it. There would not be an avalanche of laboratories from the world's most prestigious universities faithfully reproducing every detail, and if there were, any discrepancy from the "rounding" would be dismissed as "statistical fluctuation," not fraud.

This is why the present research program operates under the principle of *Sincere Science*: science that everyone can understand and build upon; science where the falsification of any theory is applauded even by those whose work was partially falsified, because the collective interest is the curiosity of discovery; empirical science about causality. It is the photoelectric effect, not General Relativity.

9. Conclusion

This paper has demonstrated that the transition from Caloric to Statistics sacrificed falsifiability for mathematical elegance. The Second Law transformed from a scientific prohibition (heat cannot flow cold $\rightarrow$ hot) to a non-scientific accommodation (heat flow cold $\rightarrow$ hot is unlikely). By Popper's criterion, theories that cannot be refuted by any

conceivable event are not science.

Three independent lines of evidence converge on this conclusion. The butter experiment dialogue demonstrates that the theory's defenses are constructed *a priori*—no experiment is needed to expose the immunization, because all outcomes are pre-explained. The distilled water experiment demonstrates that when predictions are pre-registered and data contradict them, the system deploys sequential escape routes until it must deny the validity of its own instruments. The formal logic proof demonstrates that a theory operating with "extreme improbability" rather than "impossibility" is structurally tautological—it accepts any outcome for any premise.

We propose replacing a self-protecting maze of mathematics with a simple, falsifiable, force-based framework. The thermal force hypothesis makes prohibitive predictions. The predictions are falsifiable. The Black Swans are not just possible; they are observable. Physics has built a magnificent mathematical cathedral—but on foundations of sand. The question is not whether mainstream physics has produced accurate predictions (it has). The question is whether it satisfies the definition of science itself.

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References

[1] Guzzon, A. (2026a). Kitchen Thermodynamics: Experiments 1–17. Zenodo. DOI: 10.5281/zenodo.18849653

[2] Guzzon, A. (2026b). Gravity as the Emergent Manifestation of Dual Attractive Forces. Zenodo. DOI: 10.5281/zenodo.17884248

[3] Kuhn, T.S. (1962). *The Structure of Scientific Revolutions.* University of Chicago Press.

[4] Popper, K.R. (1959). *The Logic of Scientific Discovery.* Hutchinson.

[5] Guzzon, A. (2026c). Is Mainstream Science Fictional? The 64-Combination Magnet Experiment. Zenodo.

[6] Guzzon, A. (2026d). Three-Dimensional Magnetic Asymmetry and Thermal Gradient Forces. Zenodo. DOI: 10.5281/zenodo.19162657

[7] Guzzon, A. (2026e). The Unification. Zenodo. DOI: 10.5281/zenodo.19066265

Appendices

A. Complete ChatGPT Dialogue: Butter Experiment (English)

The following is the unedited transcript of a dialogue between Dr. André Guzzon and ChatGPT (OpenAI) regarding the falsifiable predictions of statistical thermodynamics for a simple butter-on-frying-pan experiment. This dialogue constitutes primary empirical evidence of Framework Lock—the a priori immunization of a theory against any possible experimental outcome.

Dr. André Guzzon: Is statistical thermodynamics pseudoscience?

ChatGPT: No — statistical thermodynamics is not pseudoscience. It's one of the most rigorously validated frameworks in all of physics.

But let's not answer this in a superficial "authority says so" way — let's dissect why.

What statistical thermodynamics actually is

Statistical thermodynamics connects:

- Microscopic behavior (atoms, molecules, quantum states)

- Macroscopic laws (temperature, pressure, entropy)

Instead of guessing, it derives macroscopic laws from first principles using probability.

Core idea:

> If you know how particles can arrange themselves, you can predict what > systems will do.

The key object: the partition function

This is the backbone of the theory:

\[\ Z = \\sum_i e\^{-E_i / kT}\ \]

From this single expression, you can derive:

- Energy

- Entropy

- Pressure

- Heat capacity

- Phase transitions

This is not hand-waving — it's mathematically exact given the model.

Why it is NOT pseudoscience

1. Precise predictions (not vague narratives)

Statistical thermodynamics makes quantitative predictions that match experiments:

- Heat capacities of gases and solids